

The Estimator Decides:

What Curricula and Failure Recycling Can and Cannot Do in RL with Verifiable Rewards

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Code, logs, and proofs: <https://github.com/linjiw/curriculum-maxrl>

Abstract

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are treated as objective-agnostic add-ons. We show they are not: the advantage estimator underneath shapes what each can contribute, in closed form. For success-conditioned (MaxRL-style) advantages, the expected absolute coefficient mass a prompt emits from N rollouts is exactly $2(\text{pass}@N - \text{pass}@1)$ — a compute-indexed learnability functional, vanishing exactly at pass rates 0 and 1 with a single interior peak set by the rollout budget alone — and the practical drop-all-fail implementation is unbiased for the truncation-order $T=N-1$ objective, not $T=N$. Coefficient mass is a surrogate (it is not gradient norm), but it is knowable before spending a rollout, and it predicts an estimator-conditioned divergence we then measure: across nine heterogeneous maze runs, every success-conditioned run grew $\text{pass}@k$ coverage and every GRPO run lost it — consistent, exploratory evidence that motivates the balanced factorial we outline. A two-seed GSM8K pilot points the same direction (teacher-deficit sign under GRPO in 2/2 seeds; the registered regression shape in 1 of 2). The same algebra flags a failure mode of recycling that we observe and replicate at three seeds on Countdown: relabels to already-reachable destinations improve mean accuracy while *losing* $\text{pass}@k$ coverage — **recycling-induced sharpening** — and it motivates the mitigation, gating relabel admission by the destination’s estimated pass rate. At the frontier tier the moderate gate returned coverage to the no-recycling baseline while keeping most of the mean gain; the tested settings give one useful operating point, not yet a monotonicity guarantee. The boundary of the safety claim is measured too: a pre-registered fifth suite whose learnable region collapses to one template stratum defeats *every* estimator. One diagnosis covers all of it: audit the estimator before shipping a curriculum or a recycler, and measure coverage alongside mean accuracy — the currency in which every failure mode here is visible.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation can dominate wall-clock (in our committed telemetry, eight generation-dominated sessions spanning 282 steps average 83% of step time; across all 1,188 mixed GSM8K/Countdown steps, including CPU-bottlenecked maze-scale sessions, the step-weighted mean is 31%), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field’s response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [23]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator’s own algebra. The organizing fact is the shape of one function. **The estimator’s expected coefficient mass $u_N(p)$ vanishes exactly at pass rates 0 and 1 and peaks once in between, tracing three regimes of difficulty: a dead zone, a learnable region, and a mastered tail. Curricula reallocate compute within the learnable region; verified relabeling is a way to create a positive update for an all-fail requested task; and relabels that land in the mastered tail buy sharpening, not signal. The estimator shapes all three.** Three questions unpack this:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected coefficient mass a task emits from a group of N rollouts has a closed form: $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose two exact zeros anchor the difficulty regimes (§3). Mass is a surrogate for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the learnable region’s location, peak, and N -scaling come from the budget you already chose, and published learnability curricula are the $N=2$ slice. It also bounds the channel. Allocation can only redistribute signal that exists: a perfect-information allocator ties our cheap posterior in end performance (§6.1), and at LLM scale a prompt-level teacher starves before it pays (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1)\times$ more mass than RLOO on hard tasks as $p \rightarrow 0$ and releases mastered tasks; GRPO’s variance-normalized weights keep spending on both tails. We predict and measure the resulting divergence at two scales: across the audited maze runs, every MaxRL-labeled run grows pass@8 coverage and every GRPO-labeled run loses it (§6.3; exploratory — the run pool is heterogeneous); on GSM8K, the pre-registered prediction that GRPO+teacher would be the only regressing cell landed on the registered run (§6.7). Recycling has its own failure mode, and the algebra flags it: relabels that land in the mastered tail sharpen. On Countdown arithmetic this is **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@ k coverage — the data-induced sibling of GRPO’s weight-induced collapse, formally hindsight’s marginal-distribution shift [21] composed with curation-driven self-consumption [22].

Q3: Does the algebra also supply the fix? It motivates one. A relabel to a destination the model reliably reaches sits in the mastered tail, where mass vanishes, so we gate relabel admission by the destination’s estimated pass rate: the high- p zero of the same utility that schedules sampling, read as a saturation threshold on what the recycler may admit. Across three seeds the moderate gate returns frontier-tier coverage to the no-recycling baseline while keeping most of recycling’s mean gain; the available settings (moderate at three seeds, strong at one seed under corrected code) sketch a mean-versus-coverage trade but do not yet establish a monotone dial (§6.9). Where recycling is healthy the gate is transparent: hard-destination relabels pass untouched, so the validated testbeds reproduce their results with the gate enabled. One method, no per-domain switches.

Our contributions:

1. **A closed-form curriculum** (§3): the expected coefficient mass of a prompt under success-conditioned RL is exactly $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose peak location and N -scaling are derived rather than tuned, and whose exact zeros at $p \in \{0, 1\}$ anchor the dead zone, the learnable region, and the mastered tail; published learnability curricula are its $N=2$ slice.
2. **A derived teacher and a characterized recycler** (§4–5): FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract is characterized as a verified, adaptively selected auxiliary update — on-policy for the achieved task only under an explicit distribution-matching condition (Remark 3).
3. **An estimator-conditioned coverage divergence** (§6): under identical curricula, success-conditioned runs grow pass@ k coverage while GRPO runs lose it — zero exceptions across the audited maze cohort, with the same sign at LLM scale in a two-seed pilot; we grade this evidence as exploratory-but-consistent and specify the balanced factorial that would confirm it.
4. **A new failure mode and its derived mitigation** (§6.8–6.9): exact-verifier recycling improves mean accuracy while losing coverage — recycling-induced sharpening, replicated at three seeds — and gating relabels by destination pass rate yields an operating point that restores frontier-tier coverage while keeping most of the mean gain.

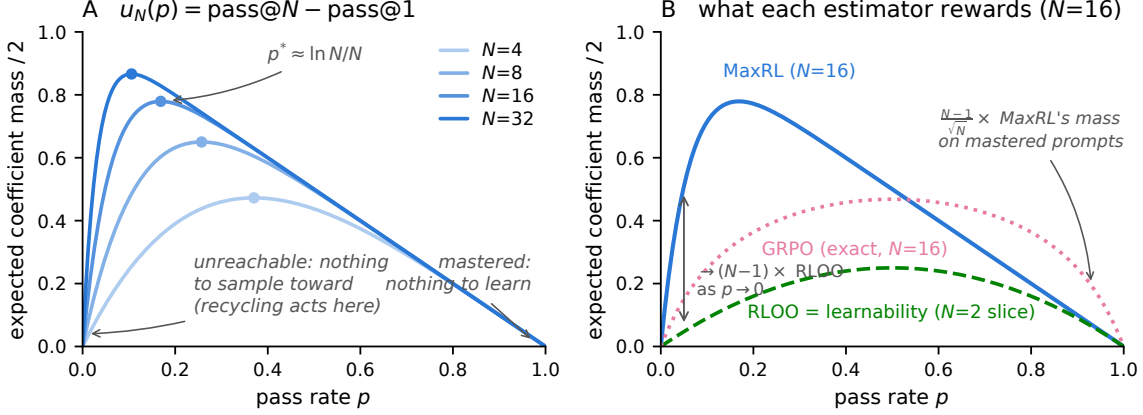


Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1 - (1 - p)^N) - p$ — half the estimator’s expected coefficient mass $A_N = 2u_N$ (Prop. 1) — for growing rollout budgets N . The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; the utility vanishes exactly at $p \in \{0, 1\}$. *Right:* all three curves are exact expected half-mass at $N=16$ for the estimators *as deployed* (GRPO with sample-SD normalization: $\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E} \sqrt{K(N-K)}$, $K \sim \text{Bin}(N, p)$; MC-verified against the code). MaxRL concentrates $\rightarrow (N-1) \times$ more mass than RLOO on frontier prompts (Prop. 2) and releases mastered prompts fastest; GRPO’s variance normalization keeps $\sim (N-1)/\sqrt{N} \times$ MaxRL’s relative mass on mastered prompts — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO’s profile is exactly the published “learnability” objective — the $N=2$ slice.

Along the way we state a refinement to the MaxRL paper — the practical drop- $K=0$ estimator is unbiased for the $T=N-1$ objective, not $T=N$ (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§8) and audited in the repository.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1 - p)$ and truncates at order T , giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T=1$) to exact ML ($T \rightarrow \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \rightarrow 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator is the curriculum

Lemma 1 (Truncation order of the practical estimator). *Let $J_T(p) = -\sum_{t=1}^T (1-p)^t/t$ denote the T -truncated ML objective, so $\nabla J_T = (\sum_{j=0}^{T-1} (1-p)^j) \nabla p$; $T \rightarrow \infty$ recovers exact ML and $T=1$ REINFORCE. Under N i.i.d. binary rewards with pass rate p , the drop- $K=0$ variant of the practical MaxRL estimator has expectation $(\frac{1-(1-p)^N}{p} - (1-p)^{N-1}) \nabla p = (\sum_{j=0}^{N-2} (1-p)^j) \nabla p$: it is exactly unbiased for J_{N-1} , not J_N , for $N \geq 2$. Zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing the $-(1-p)^{N-1} \nabla p$ correction, and $\sum_{k=1}^N (1-p)^{k-1} - (1-p)^{N-1} = \sum_{k=1}^{N-1} (1-p)^{k-1}$.*

Interpretation. A refinement to Tajwar et al. [1], immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped. Magnitude at the deployed budgets: the relative gradient shortfall of $T=N-1$ vs. $T=N$ is $(1-p)^{N-1} / \sum_{k=1}^N (1-p)^{k-1}$ — at $N=16$ it is 5.8% at $p=.01$, 1.1% at the band peak $p^* \approx .17$, and $< 10^{-4}$ at $p=.5$; at $N=8$ it reaches 12% on the deep frontier. Bookkeeping at moderate p , material exactly where the frontier teacher concentrates sampling.

Proposition 1 (Expected coefficient mass). *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights, define the utility $u_N(p) := \text{pass@}N(p) - \text{pass@}1(p) = (1 - (1 - p)^N) - p$. Then the expected absolute coefficient mass is*

$$A_N(p) := \mathbb{E}\left[\sum_i |w_i|\right] = 2(\text{pass@}N(p) - \text{pass@}1(p)) = 2u_N(p). \quad (1)$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1 - \frac{K}{N})$, and $\mathbb{E}[2(1 - \frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(\text{pass@}N - \text{pass@}1)$. The sampling utility used everywhere below (figures, Algorithm 1, code) is u_N ; the factor two matters only when quoting exact mass, and we carry it explicitly as $A_N = 2u_N$. Two exact companions tie mass to the gradient. Factorization: writing $\mu_{\pm}(x) = \mathbb{E}[S \mid r=1/0, x]$ for the success/failure-conditioned score means, exchangeability gives, for every group size,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p) (\mu_+(x) - \mu_-(x)), \quad (2)$$

so u_N is exactly the estimator-controlled scalar multiplying the prompt’s success–failure score contrast — the part of the expected update a sampler can know before rolling out; the contrast (and where it transfers) is the part it cannot. Weight view: since $\mu_+ - \mu_- = \nabla p/(p(1-p))$, Eq. (2) is equivalent to $u_N(p) = p(1-p) w_{N-1}(p)$ with $w_T(p) = (1 - (1-p)^T)/p$ the truncated weight function of Lemma 1 — mass and the practical estimator’s weight view are linked, not identical objects. (Both identities MC-verified; artifact `verify_guidance_math.json`.)

Interpretation. *The estimator’s expected coefficient mass on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. Coefficient mass is a surrogate for learning signal (Remark 1), but its two zeros are exact, and they anchor three regimes of difficulty: a dead zone at $p = 0$ (nothing to condition on; under the current policy and support, sampling alone cannot produce a success there — and at small $p > 0$ a task is operationally dead at budget (B, N) when its live-group probability $1 - (1 - \ell_N(p))^B$, with $\ell_N(p) = 1 - p^N - (1 - p)^N$ and B its share of groups, is too small to sustain training: on the frontier-heavy pool of §6.2 ($p \leq 10^{-5}$, ~ 133 groups per task) that probability is $\leq 2\%$ per task, and the one live group observed across all uniform seeds did not ignite — dead in the budgeted sense, not by fiat), a learnable region with a unique interior peak at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ (unique because $u_N'' = -N(N-1)(1-p)^{N-2} < 0$), and a mastered tail at $p = 1$ (no contrast left). The regimes are continuous — the only exact zeros are $p \in \{0, 1\}$ — so where we quote a “band” we mean an explicit threshold set $\{p : u_N(p) \geq \eta u_N(p^*)\}$ ($\eta=0.5$ unless stated), whose location and N -scaling follow from the one parameter you already chose. Everything in this paper is one of three moves against this structure: allocate within the learnable region (§4), create signal at its bottom (§5), and refuse updates that land in the mastered tail (the gate).*

Proposition 2 (Frontier amplification). *As $p \rightarrow 0$, the ratio of MaxRL’s expected mass to RLOO’s ($A_N^{\text{RLOO}} = 2p(1-p)$ exactly, under the $1/N$ -normalized leave-one-out weights) tends to $N - 1$.*

Interpretation. *The comparison to GRPO is two-sided and exact for the deployed implementation, which normalizes by the **sample** standard deviation ($s_K = \sqrt{K(N-K)/(N(N-1))}$, PyTorch’s default), not the population SD: $A_N^{\text{GRPO}} = 2\sqrt{\frac{N-1}{N}} \frac{1}{N} \mathbb{E}\sqrt{K(N-K)}$ with $K \sim \text{Bin}(N, p)$ (MC-verified against `estimators.py`). Taking tails, MaxRL/GRPO $\rightarrow \sqrt{N}$ as $p \rightarrow 0$ while GRPO/MaxRL $\rightarrow (N-1)/\sqrt{N}$ as $p \rightarrow 1$ (at $N=16$: $4.0\times$ and $3.75\times$; the symmetric $\sqrt{N-1}$ holds only under population-SD normalization). MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sim (N-1)/\sqrt{N} \times$ more of its mass on mastered prompts — maintenance a curriculum then removes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.*

Proposition 3 (Allocation, static). *Consider the static problem $\max_{\{N_i\}} \sum_i u_{N_i}(p_i)$ subject to $\sum_i N_i = B$ and integer $N_i \geq 0$, with pass rates p_i fixed and known and one group per prompt. Because the marginal gain $u_{N+1}(p) - u_N(p) = p(1-p)^N$ is decreasing in N , greedy water-filling on this marginal — the probability that the next rollout is a group’s first success — is optimal.*

Interpretation. *This is a mass-optimal reference allocator for the static, known- p problem only; it does not prove that sampling prompts $\propto u^\gamma$ is optimal (at fixed group size, expected immediate mass is linear in the sampling distribution, so the unconstrained myopic optimum is a brittle arg-max), nor does it describe the §6.1 oracle, which samples prompts by true- p utility at fixed group size rather than varying N_i . The practical teacher (§4) keeps fixed group sizes and samples prompts $\propto u^\gamma$ with a floor, trading myopic optimality for group-parallel batching and exploration. We measure what the gap costs: a **true-pass-rate utility oracle** (no floor, γ -matched) ties the Thompson teacher in end performance (§6.1).*

Remark 1 (Scope of the surrogate). Coefficient mass counts weighted rollouts, not score-vector geometry. We use it as the sampling utility because (i) it is knowable *before* spending rollouts, (ii) its zeros are exact — at $p \in \{0, 1\}$ the estimator emits no gradient of any norm — and (iii) on the exact-gradient testbed it *ties* the true first-order expected improvement as a task ranking (Spearman within .005 at every snapshot, and exactly 1.0 against the full exact transfer matrix — one expected group per task, exact Δp on every other task — at three training snapshots; artifacts `results_bridge.json`, `results_transfer_matrix.json`). That tie is a property of *shared-prefix* task structure, where activity and transfer align; on pools without it, activity ranks what a task can emit, not where it helps (§6.5–6.6 measure that gap). What it does not claim to be is *the* optimal sequential utility, and no state-local functional is: on independent tasks the true final-performance optimum is a planning solution (trajectory-knapsack with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^{\alpha} u_N$ wins anytime performance while model-predictive replanning wins the final checkpoint, each losing the other’s metric (8–10/10 paired seeds each way; App. A). “Best curriculum utility” is ill-posed until that objective is fixed; we report both currencies throughout. The load-bearing claim is the pair of exact zeros at $p \in \{0, 1\}$, which every candidate utility shares.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x \sim q_t}[J(x)]$: each group’s gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ . This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t = \rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ , so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ -measured outcome.

One identity, three literatures: learnability curricula [11, 29] are the $N=2$ slice of u_N ; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [15] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). DisCO [2] and SC-SDPO [3] derive analogous per-question weights for the GRPO family to diagnose difficulty bias; our identity generalizes that line to success-conditioned estimators and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The three regimes, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the learnable region — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal where sampling the requested task yields none — in our control battery, the one intervention that still adds on top of the oracle. The **estimator** underneath shapes whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe*.

4 FrontierMax: the schedule

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior tracks each prompt’s pass rate from observed group outcomes only — never from relabeled successes, which inflate it ($\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^{\gamma}$ with a uniform floor. The knobs (decay 0.7, floor 0.1, $\gamma \in \{1, 4\}$) carry validated defaults (Table 3); what is *derived* rather than tuned is the difficulty band itself — location, width, and N -scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Remark 3 (Hindsight update, characterized). Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields a *verified weighted-likelihood update for the achieved task*: every relabeled example is a true success of g' under the environment’s verifier. It is generally an *adaptively selected, off-policy auxiliary update*, not an

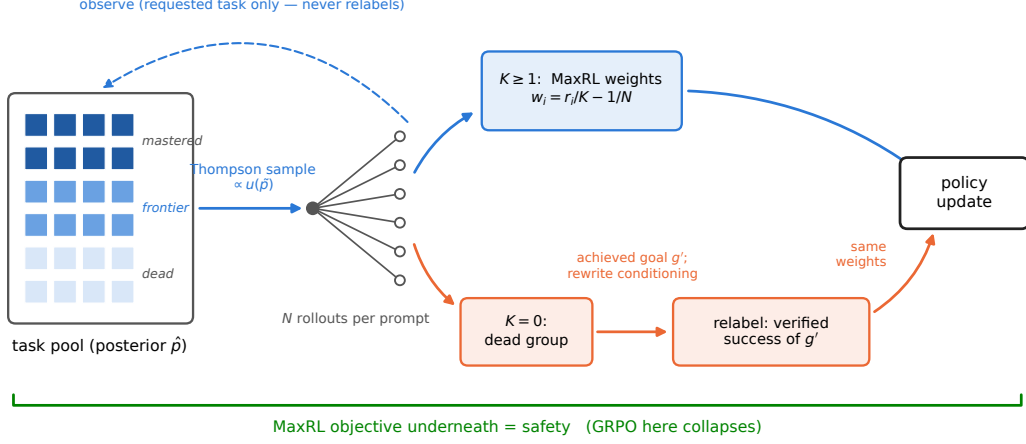


Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked ▷).

- 1: ▷ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \text{Beta}(\alpha_x, \beta_x)$ (else: uniform)
- 2: roll out N completions per prompt; verify; $K_x \leftarrow$ successes
- 3: update posterior (α_x, β_x) from (K_x, N) *for the requested task only*
- 4: live groups ($1 \leq K_x$): MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
- 5: ▷ **dead groups** ($K_x=0$): relabel each parseable failure to *its own* achieved sub-goal g' (verified; conditioning rewritten); unparseable rollouts stay failures, so the group retains contrast ($K < N$)
- 6: ▷ admit the relabel only if $\hat{p}_{g'} \leq \text{gate_max_p}$ (the gate: refuse the mastered tail; never-seen destinations start at the Beta(1,1) prior $\hat{p} = .5$, admitted at the default threshold)
- 7: one policy-gradient step on live + admitted relabeled groups

unbiased gradient of g' 's on-policy objective: g' is selected from the same group being scored, conditioning on the requested task's failure correlates the rollouts, the conditioning rewrite changes the law under which the log-probability is evaluated, no importance ratio corrects for it, and the finite- N weights target J_{N-1} (Lemma 1) rather than exact ML. It equals the fresh-rollout J_{N-1} gradient for g' only under a distribution-matching condition — the selected, rewritten group must have the same joint law as an i.i.d. fresh rollout group for g' . This specializes known relabeling-bias structure — the induced-distribution view of Eysenbach et al. [20] and the conditional-law gap in Ghosh et al. [21] — to success-conditioned RLVR weights; we state it as a remark, not a result, and note that the sharpening phenomenon of §6.8 runs through the *marginal* shift those works also identify, not this conditional matching.

Interpretation. “Exactness” is a property to measure per environment, not a blanket guarantee. On our validation testbed the laws match by construction, so cosine probes there are consistency checks, not evidence (several null relabeling schemes produce identical cosines); the evidence that the relabel direction carries information is behavioral — the placebo decomposition below. We measured the two gaps the remark names where they are measurable (exact chains, 5 seeds; artifact `results_proposal_shift.json`): the applied relabel update’s cosine against the exact fresh-destination gradient direction is 0.93, the relabeled group’s success rate exceeds the true fresh destination pass rate by only +.003 (selection guarantees $K' \geq 1$, so the bias exists but is small here), and cross-fitting the destination choice (select on half the group, update with the other half) removes the adaptive reuse at a real cost: alignment drops to 0.82 and AUC falls .883 $\rightarrow$.829, because the selected successes leave the update half. The off-policy caveat is real; its sign is environment-specific.

Two contracts keep practice on the proof’s side: (1) **exactness** — a relabeled success must be a true success of the relabeled task under the environment’s own verifier (never an LLM judge); (2) **conditioning**

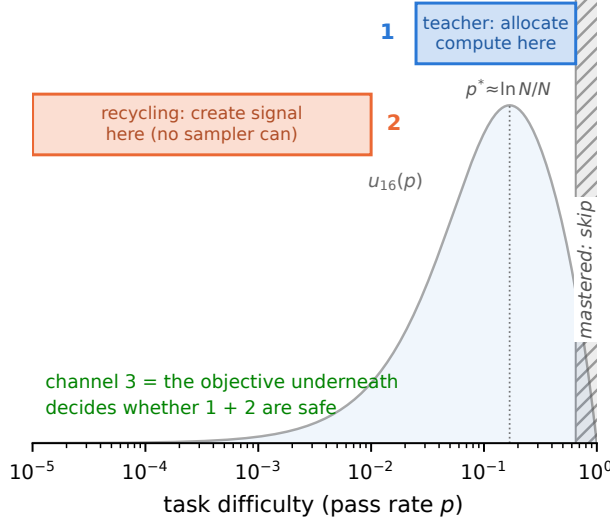


Figure 3: **The three regimes as a map of the method.** The teacher allocates compute within the learnable region; recycling creates signal where sampling the requested task yields none; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

— goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Among the interventions studied here, recycling is the one that *creates* signal for an all-fail requested task (adding the achieved sub-goals to the pool by hand would too — what recycling buys is discovering them, §6.2), and we decompose its headline gain rather than let it inflate: of the $+0.22$ AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries $+0.037$ beyond the strongest placebo, is the tightest arm in the battery, and is exactness-dependent (random-target relabels lose to dose-matched replay 5/5). Both numbers ship together. The value is regime-dependent in a way we can state precisely: $+0.22$ AUC on fixed task pools, $+0.01$ on one-shot task streams — the gain is proportional to how much a relabeled skill can compound, and task-set fixedness is the single most important regime variable we found.

6 Experiments: an escalating ladder

Each rung asks one question of the three-regime structure, at increasing scale and decreasing experimental control; Table 1 is the map. Statistical contract, stated once and used throughout: the independent unit for any training-method claim is an independently trained seed block; runs sharing a seed, warmstart, or checkpoint are paired or correlated observations, never additional replicates; levels, k values, metrics, and checkpoints from one trained model are repeated measurements of it; repeated-evaluation spread measures evaluation noise, not training-seed uncertainty, and every $\pm$ names its source at point of use ($\pm$ with no qualifier is standard deviation across training seeds). “Coverage” is pass@ k with exactly k samples drawn per prompt and averaged over the eval pool, at the k named per suite (8 on the maze, 16 on Countdown, 4 on GSM8K); mean@ k is the mean success fraction of the same k samples; AUC is the mean of the eval curve over training at matched wall-clock or episode budget, and final is the last checkpoint. Where run pools are heterogeneous we say so and report signs descriptively rather than attaching p -values (§6.3–6.4). Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository’s audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform $0.650 \rightarrow$ teacher $0.728 \rightarrow$ full stack 0.890 ± 0.002 AUC. An eight-arm control battery (App. A) orders into a

Table 1: The ladder at a glance: every rung asks one question of the three-regime structure; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3	main effect, real grads?	maze transformer, 1.26M	9 runs	9/9 signs (exploratory)
6.4	where/when does coverage go?	maze, level-resolved	22 runs	easy band, immediately
6.5	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6	beyond task spread?	MountainCar/CartPole	3	spread=existence; tea.=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	sign landed; replication open
6.8	what does recycling cost?	Countdown v2	3	mean up, coverage down
6.9	does the gate recover it?	Countdown v2	3	one useful trade point

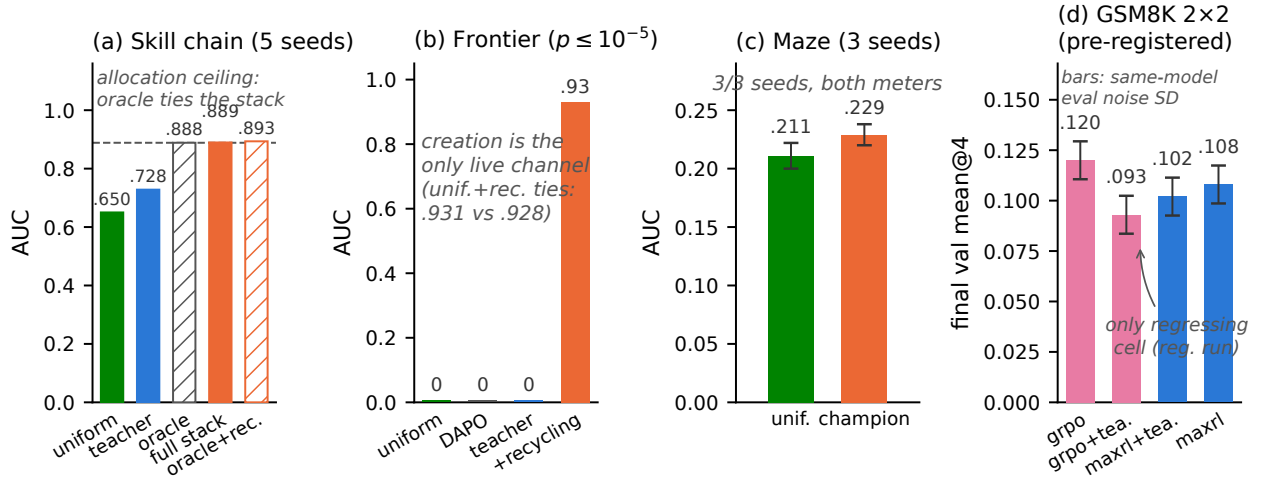


Figure 4: **The ladder.** (a) Exact-gradient chains: a no-floor, γ -matched true-pass-rate oracle matches the full stack within the observed five-seed variation — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 3/3 seeds on both meters. (d) GSM8K 2×2, registered run, error bars = measured same-model eval noise (SD .0094): GRPO+teacher is the only cell that regresses (pre-registered P-G2; a replication seed did not reproduce it — §6.7); the teacher’s MaxRL effect is a null at this budget (P-G1).

gradient-information ladder, each rung a 5/5 paired-seed win: zero-information dose (0.798–0.800) < replayed correct direction (0.832) < true relabels (0.869) < perfect allocation (oracle 0.8885 $\approx$ full stack 0.8895) < allocation+creation (0.8935). Two decompositions matter downstream. *Allocation saturates*: a no-floor, γ -matched true-pass-rate oracle (0.8885 $\pm$.0014) sits within the five-seed variation of the full stack (0.8895 $\pm$.0023) — similar within observed noise, not tested for equivalence — so perfect difficulty information is worth little over the cheap posterior. *Creation decomposes*: a replay placebo captures 83% of recycling’s +0.22; the relabel *direction* carries +0.037 beyond it and is the most reliable arm in the battery, while random-target relabels lose to dose-matched replay 5/5. The utility form is load-bearing through its N : the N -blind learnability slice $p(1-p)$ forfeits the entire allocation gain (−.007 vs. uniform) where u_N carries +.078 (5/5). All arms converge within .027 of one another — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling.

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; every intervention buys speed, not asymptote.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5} ; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ~ 400 groups, then goes silent. The attribution control pins down which channel did it: *uniform*+recycling scores the same ($0.931 \pm .002$ vs. teacher+recycling $0.928 \pm .003$ AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy’s own failures, with no task engineering. One loss-level alternative remains: the MaxRL paper’s own Eq. (10) keeps the control-variate term $-\frac{1}{N} \sum_i S_i$ on all-fail groups, so that estimator is *not* silent at $K=0$ ($\mathbb{E}[\hat{g} \mid K=0] = \nabla p / (1-p)$, MC-verified). We ran it (5 seeds, matched budget; artifact `results_fullcv_baseline.json`): on this pool the full-CV variant spends 100% of its update norm on all-fail groups and still scores 0.000 in every seed, with or without the teacher — the all-fail term pushes down every sampled action symmetrically and cannot ignite a pool this deep, while recycling under either variant reaches 0.98. On the balanced pool the two variants tie (paired Δ AUC straddles zero, 5 seeds).

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Even the estimator family’s nonzero all-fail update (full control variate) stays at zero here: creation is the only channel we tested that ignites a dead pool, and its value is discovery, not allocation.

6.3 Does the estimator main effect survive real gradients? (maze transformer at matched wall-clock; 1.26M params, 13 levels, ~ 30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252 ± 0.005 final / 0.229 ± 0.009 AUC vs. uniform 0.230 ± 0.015 / 0.211 ± 0.011 ; paired deltas positive in 3/3 seeds on both meters (final and AUC — six correlated contrasts, not six seeds); 22–35% more optimization steps per GPU-hour. Attribution within that gain, stated plainly: matched *wall-clock* lets curriculum arms take 1.2–1.4 $\times$ more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher’s effect (step-matched teacher-only Δ AUC +0.006, within seed noise) while teacher+recycling survives step-matching (positive in all three paired seed blocks; a paired t at $n=3$ gives $p \approx .04$, which we report as descriptive given its normality sensitivity — an exact sign-flip test cannot go below .25 at three blocks) — on the maze the teacher’s measured value is *throughput* (buying more steps per GPU-hour), recycling’s is signal. The maze teacher uses the legacy frontier form; a pre-registered 3-seed rerun with the exact u_N measures the two within noise *on the maze itself* (AUC $.2138 \pm .0059$ vs. $.2135 \pm .0104$; artifact `un_form_verdicts.json`), closing the gap between the deployed form and the derived one. The same sweep refutes a CPU-derived within-band refinement (a $(1-p)$ -tilted utility that won 10/10 on the testbed loses 0/3 on the maze on every meter) — within-band utility choices are testbed-sensitive, which is why the theory claims only the band’s boundaries. Safety, stated as what the design supports: an estimator-associated coverage difference — all five MaxRL-labeled runs grow pass@8 (e.g. $0.316 \rightarrow 0.348$ under the teacher) and all four GRPO-labeled runs lose it ($0.308 \rightarrow 0.271$ uniform; $0.332 \rightarrow 0.269$ with the teacher). We state this as what it is: perfect sign separation across nine runs whose pool mixes sampler configurations (the MaxRL five span frontier and uniform teachers; the GRPO four span uniform and one frontier) and reuses seed blocks and warmstarts, so the runs are not exchangeable and a label-permutation p -value would answer a randomization this design did not perform — we therefore report zero exceptions descriptively and do not attach one. Two sensitivity reads support the same direction without curing the design (artifact `permutation_reanalysis.json`): restricting permutations within matched sampler arms preserves the perfect separation, and all three shared-warmstart pairs are individually positive; the run composition (5v4 because one uniform-MaxRL seed predates pass@ k logging) is tabulated in the same artifact. The confirmatory design this motivates is a balanced $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, \text{teacher}\}$ factorial over at least six independent seed blocks with identical warmstarts and one prespecified endpoint — fewer than six blocks cannot reach two-sided $p < .05$ even with unanimous signs. At the exact-gradient rung, where the design is cheap, we already ran its schedule-matched core: freeze each seed’s *realized* prompt sequence (one recorded teacher trajectory, one uniform), then replay the identical sequence under each estimator from the same init — the estimator ordering on exact summed pass@8 survives with the schedule held fixed (MaxRL over GRPO in 10/10 seed $\times$ schedule pairs; artifact `results_schedule_matched.json`), with the

caveat that on that learnable-everywhere pool GRPO still gains coverage — the neural-scale *loss* is what the balanced maze factorial must test. Whether the teacher *amplifies* GRPO’s decay is a factorial (interaction) question this suite is not powered for; the GSM8K 2×2 below tests exactly that cell, pre-registered. Inference currency (one convention, every level: smallest k reaching absolute coverage 0.25; single checkpoint pair): our checkpoint needs 11× fewer samples at level 5, ties GRPO at levels 2–3, and needs 3× more at level 4 — the frontier is where the efficiency gain lives. The summed-coverage view agrees without a target choice: GRPO ahead at pass@1, curves crossing at $k \approx 4$, MaxRL ahead by half a level at $k=64$ (App. A) — choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO’s coverage go? (level-resolved). Decomposing the main effect by difficulty band turns it into a mechanism (Fig. 5). *Where:* GRPO’s pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 (−.224 mean over its four runs, every run negative) while its L1–3 pass@1 *rises* (+.02 to +.14, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat (+.005±.058 over 18 runs) and concentrates its gain on the frontier band L4–6 (+.126±.063 vs. GRPO’s +.057). The 18-run MaxRL pool is a heterogeneous run inventory (teacher variants, hindsight doses, a wide model, repeated seeds), so we report the band asymmetry descriptively — every GRPO run negative on L1–3, the MaxRL spread centered at zero there — and attach no permutation p -value; the 22 runs are not exchangeable replicates of one estimator contrast. This is consistent with the mass curves in Fig. 1 — the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts where u_N has already released them — but the step from mass to coverage is a stated *hypothesis*, not a derivation: mass is sign-blind, and “mastered-tail updates sharpen” requires an argument about the sign structure of near-saturated groups (a lone failure draws a large negative weight under variance normalization, pushing mass off the failure mode and onto the dominant one) that we support with the location and timing evidence here rather than prove. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total diversity-premium loss (pass@8 − pass@1: .123 → .042) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only .62–.75; MaxRL’s premium stabilizes after warmup (.122 → .101). A dose-response note from the same logs (descriptive, same caveat as above): sparse relabeling (1/dead group) uniquely preserves the diversity premium relative to none and dense, while dense banks the same coverage into the best final pass@1 — dose sets *where* the coverage dividend is spent, not whether coverage is lost.

Takeaway. The gains survive real gradients, and the coverage main effect now has a located mechanism: GRPO liquidates easy-band coverage for reliability from the first updates; MaxRL reallocates the same budget to the frontier. Coverage is the currency that sees it.

6.5 When does allocation *not* pay? (IsaacLab legged locomotion, pre-registered null). A 5-arm Anymal-C rough-terrain pilot (one seed, run by an independent team on a co-designed protocol; App. A) is consistent with its pre-registered null — underpowered alone at one seed, weight-bearing only jointly with the maze and Countdown regime evidence: on a terrain grid where every level is learnable at budget, the stock greedy curriculum beats the frontier teacher outright (mean pass .372 vs. .278). Read with the maze step-matched analysis and Countdown, the consistent regime pattern — at one seed here, so a pilot-level observation rather than proof — is: **allocation pays where unlearnable-at-budget regions exist to avoid**; on learnable-everywhere pools, breadth beat focus in every case we measured.

6.6 What does a curriculum add beyond task spread? (Gymnasium control — binary-success RLVR on classic control; official-task currency). All arms train on *binary task success* (MountainCar: reach the flag; CartPole: survive 400) — the classic-control analogue of a verifier reward. The **target-only** arm is then a provable zero-update freeze, not a slow learner: every group is all-fail, every group-relative estimator emits exactly zero weights on constant rewards, and telemetry confirms zero parameter updates over the full budget (random-policy success $< 3 \times 10^{-6}$; no budget ignites it). The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3 seeds, Fig. 8). A dense-reward control confirms the zero is a property of sparse verifier-style reward (REINFORCE on CartPole’s native reward solves it; App. A). Attribution at a common horizon: uniform-over-bins task spread supplies most of the effect (existence of a gradient), recycling-without-allocation plateaus below ceiling, and the teacher’s remaining

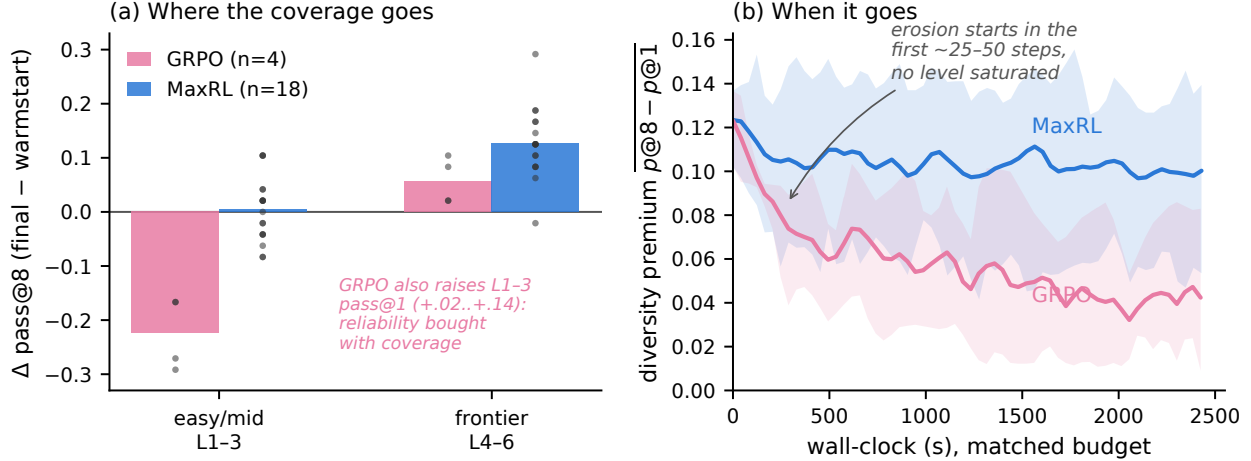


Figure 5: **Where and when GRPO’s coverage goes (maze, level-resolved, matched wall-clock).** (a) $\Delta \text{pass@8}$ (final – warmstart) by difficulty band: GRPO’s loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; a heterogeneous run inventory — descriptive, no p -value). (b) Diversity premium $\text{pass@8} - \text{pass@1}$ vs. wall-clock (mean and min–max across runs): GRPO’s erosion begins within the first ~25–50 steps, before any level saturates; MaxRL stabilizes after warmup.

contribution is speed and stability ($3\times$ faster to hard-task 0.9, $2.5\text{--}6\times$ smaller cross-seed variance). A methods note we keep because it generalizes: our own first version used per-bin task-partitioned policy parameters and reproduced the known failure (0.000 in every arm) — curricula operate through shared parameters, or not at all.

Takeaway. Curricula operate through shared parameters, or not at all.

6.7 Does it transfer to LLM scale? (GSM8K 2×2 , pre-registered; SmolLM2-360M, $N=16$, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half, and the deficit against plain GRPO at the endpoint is $2\text{--}4\times$ the measured same-model eval noise (mean@4 $-.027$; pass@4 $-.048$; the second-half dip itself, $.096 \rightarrow .093$, is within the $\pm .009$ noise floor and carries no weight on its own) while plain GRPO climbed monotonically ($.078 \rightarrow .105 \rightarrow .120$). One uncertainty caveat governs every GSM8K number: the $\pm .009/\pm .017$ bars are repeated-eval noise of a fixed checkpoint — they bound evaluation variability, not training-seed variability — so deficit-to-noise ratios are descriptive signal-to-noise reads, not hypothesis tests; training-seed uncertainty at $n=2$ is what the queued multi-seed cell measures. Steering telemetry, read against its own controls: the teacher’s sampled-dead fraction *minimum* reaches .48 vs. uniform’s .52, but the run means are indistinguishable (.66–.67 in all cells) — the delivered treatment was weak, which is the posterior-starvation mechanism below, and it bounds how much of the GRPO gap the curriculum can own. **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically ($.073 \rightarrow .095 \rightarrow .118$), and its steering telemetry shows the treatment was weaker (min dead-sampled fraction .53, mean .66 $\approx$ population — a near-uniform teacher on average). The completed *within-seed* pair (grpo-uniform seed 2: $.081 \rightarrow .085 \rightarrow .125$, final pass@4 .229; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both meters (mean@4 and pass@4 are computed from the same generations and strongly correlated, so this is two seed-level confirmations read on two coupled meters, not four; seed 2: $-.007$ mean@4, within noise, $-.026$ pass@4, $\sim 1.5\times$ its floor) with magnitude tracking steering intensity — a dose effect, not a sign flip. The pre-registered *second-half-regression* shape remains 1 of 2 seeds: suggestive, not established. MaxRL+teacher trains stably to its best value ($.066 \rightarrow .102$); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for

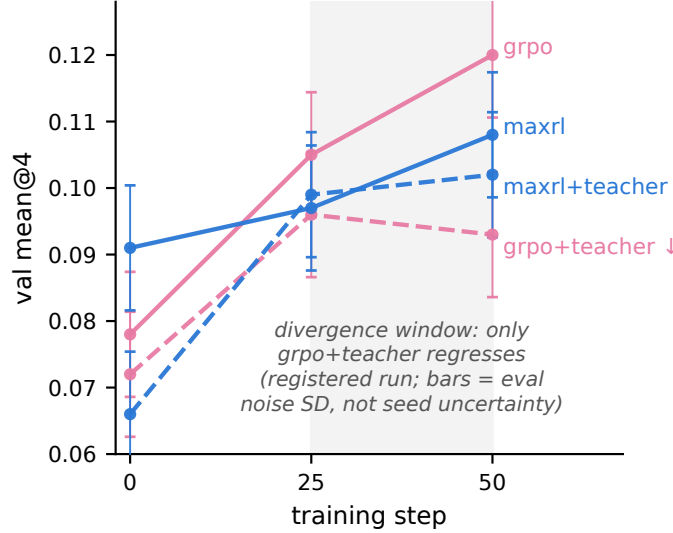


Figure 6: **GSM8K 2×2 (pre-registered; registered run shown; error bars = measured same-model eval noise, SD .0094)**. GRPO+teacher is the only cell that regresses in its second half (P-G2 on this run; the load-bearing statistic is the endpoint gap vs. plain GRPO, 2–4× the repeated-eval noise — a descriptive signal-to-noise read, not a test; the second-half dip alone is within noise; a replication seed with weaker measured steering climbed instead — §6.7); under MaxRL the identical teacher trains stably to its best value but does not beat uniform at this budget (P-G1: null).

MaxRL) a *null at this budget* — predicted in advance by the posterior-starvation analysis below. A $\text{pass}@k$ sweep on the registered checkpoints: the teacher’s deficit under GRPO *widens monotonically with k* (−.008 at $k=1$ to −.024 at $k=16$) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise (harness caveats in App. A). One design confound is disclosed rather than dismissed: the teacher samples with replacement, so curriculum cells revisit prompts more — a repetition effect is an alternative reading that a uniform-with-replacement control cell separates; it is running with the steering-controlled replication. The mechanism behind the weak treatment: the teacher’s posterior *learns* real difficulty from ~ 1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$; cross-run posterior agreement 3× its agreement with the external key — it learns *this model’s* landscape) but its allocation is *posterior-starved*: 3,200 draws over 7,473 prompts leave the estimated utility landscape nearly flat, so Thompson sampling stays near-uniform (full posterior diagnostics in App. A). Small model; the pre-registered outcomes were orderings and signs on the registered run.

Takeaway. The maze sign pattern stands on nine heterogeneous runs; the LLM-scale interaction is 1-of-2-seeds and tracks steering intensity — the decisive experiment is a steering-controlled multi-seed cell, queued.

6.8 What does recycling cost? (Countdown 2×2, pre-registered): two nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool’s simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §6.3’s weight-induced collapse — data-induced rather than weight-induced. The underlying concentration pathology is known in goal-conditioned RL (Skew-Fit’s motivating failure mode; GCSL’s marginal shift); what is unreported, to our knowledge, is its measurement in RLVR — no prior work quantifies the $\text{pass}@k$ cost of hindsight relabeling in a verifier loop. Our algebra flags the mechanism: the achieved-goal marginal overweights

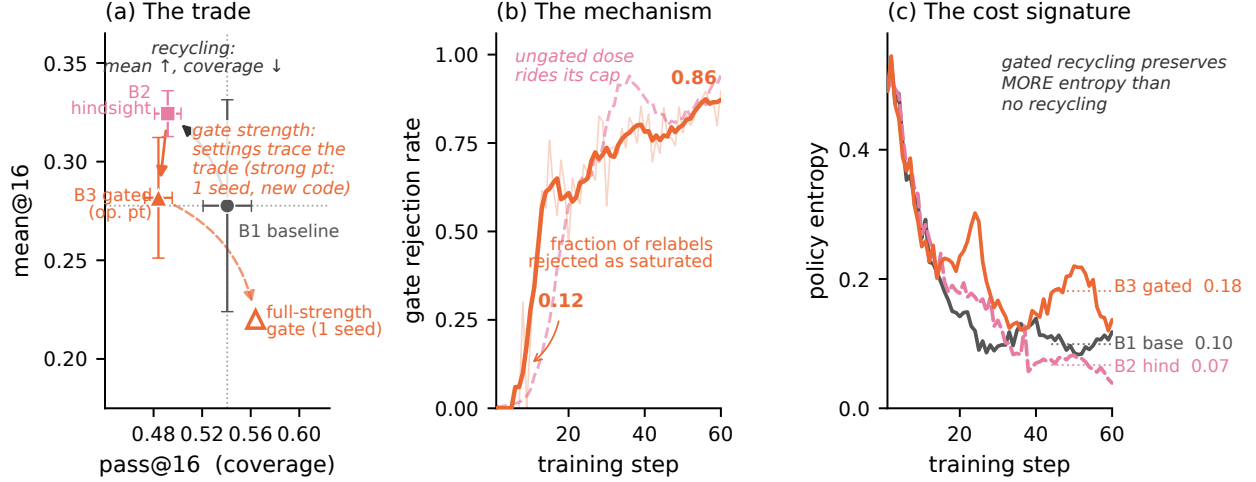


Figure 7: **Recycling-induced sharpening and its derived mitigation (Countdown, tier 1).** (a) Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage. On this tier the moderate gate (B3) gives back most of the mean without moving coverage — tier-1 destinations saturate early, so admitted relabels are few — while full-strength gating (corrected decay, 1 seed) returns coverage *above* baseline at the mean’s full price. The settings are consistent with a mean-vs-coverage trade (not yet a monotone dial: the strong point is one seed on different code), and the useful middle lives on the frontier tier, where the same moderate setting restores coverage to baseline (.279±.019 vs. .274) keeping ~60% of the mean gain (§6.8). (b) The mechanism in-run: the gate’s rejection rate rises monotonically as the model’s reachable set saturates its relabel destinations, while the ungated dose rides its cap. (c) Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

destinations the policy already reaches, so relabel updates concentrate toward the mastered tail, where $u(p)$ says fresh training signal vanishes — the softmax pays for those updates out of the exploration tail. (One selected success does not by itself imply the destination’s fresh-rollout pass rate is high; that is what the gate’s independent destination estimate $\hat{p}_g$ checks.) *Takeaway: recycling needs an admission rule, and the utility’s high- p zero motivates one — a destination-saturation threshold on the estimated pass rate $\hat{p}_g$ (the implementation thresholds $\hat{p}$ directly; it does not evaluate u_N).*

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms $\times$ three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). Setup note with a theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures (relabel yield 0.000, $n=384$), so there is nothing to recycle until SFT installs the output format; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant ~75% of the independence bound $1 - (1 - \text{pass@1})^{16}$ (ratios .744/.783/.755 across tiers) — the baseline our headroom gate measures against. **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 $\rightarrow$.324±.012) and pays -.049 of pass@16 (.541±.020 $\rightarrow$.492±.011). The coverage loss clears both arms’ seed spreads ($> 2\times$); the mean gain clears the treated arm’s tight spread but sits within the noisier baseline’s — we report it as directionally consistent across seeds rather than individually significant. The frontier tier moves the same direction. **The timing is dose-then-damage:** recycling’s frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ~28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all seeds negative*, after ~65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed; the gate’s entropy retention

replicates across seeds while ungated recycling makes entropy seed-unstable (details in App. A). **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline ($.279 \pm .019$ vs. $.274 \pm .015$) while keeping $\sim 60\%$ of the mean gain; at tier 1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely. Implementation disclosure: a decay-math bug made all three-seed gated runs gate *weaker* than designed (effective destination threshold $\hat{p} \approx 0.64$ where the design implies 0.92), so the reported operating point is an under-gated one; the corrected-decay rerun (1 seed) gates $\sim 3\times$ harder and pushes tier-1 coverage *above* baseline at the mean’s full price (Fig. 7a). The three settings are *consistent with* a mean-vs-coverage trade — no gate: full mean gain, coverage lost; under-gated: $\sim 60\%$ of the gain, frontier coverage restored; full-strength: gain erased, coverage above baseline — but they do not establish a monotone dose-response: the strong point is one seed, gate setting is confounded with the decay-code version, and at tier 1 the moderate gate moved *both* point estimates below the ungated arm. We claim the moderate frontier-tier setting as one useful operating point and defer the dial claim to the pre-registered fixed-decay sweep. In-run telemetry shows the mechanism (Fig. 7b–c): rejection rises $.12 \rightarrow .85$ as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline. A pre-registered `gate_max_p` dose sweep at fixed decay is the standing follow-up.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the derived gate buys back frontier-tier coverage at a measured mean price — one validated operating point, with the monotone-dial question deferred to a fixed-code sweep.

7 Related work

The coverage tension. RLVR itself trades coverage for precision: `pass@1` rises while `pass@k` at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator’s implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator. The estimator-difficulty interaction itself is a heating 2026 theme — SC-SDPO [3] shows GRPO’s normalization implicitly encodes a difficulty weighting, and A-GRAE [28] redesigns the advantage around difficulty focus — but no prior work runs *identical* curricula under two estimators or measures the resulting `pass@k` sign flip.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1 - \hat{p})$, which Prop. 1’s corollary identifies as the $N=2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and Prop. 3 is the idealized form of both (water-filling spends its next rollout wherever $p(1 - p)^N$ is largest, across or within). DAPO’s dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward `pass@k` but keep uniform sampling. **Goal selection in HER.** Selecting *which* goals to train on by difficulty or reachability is a decade-old idea in the relabeling literature: intermediate-difficulty goal generation [16], value-based hindsight goal selection (HGG; 17), curriculum-guided HER’s proximity-diversity admission of relabel goals [18], and Skew-Fit’s α as a concentration-vs-coverage dial [19]. Our gate belongs to this mechanism class; what is new is not admission-by-difficulty but its *derivation* — the threshold is the mastered-tail zero of the same estimator functional u_N that schedules sampling, so teacher and gate are one object with no separately tuned goal-selection heuristic — and the coverage accounting it is validated against.

Hindsight and self-training for LLMs. Exact-verifier relabeling has precedent outside RLVR loops — CodeIt [24] (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling [25] — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH [23] (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL’s hindsight response replay [26] (densifying GRPO signal for multi-turn agents). None of these measures `pass@k` under

relabeling — precisely the currency where we find recycling’s hidden cost. The nearest phenomenon report is in same-problem rejection finetuning: Strozzi [27] document a STaR-trained model winning pass@8 while the base model overtakes at pass@256 — concentration over reach — for SFT on a model’s own verified outputs. Recycling-induced sharpening is the cross-task, on-policy-RL sibling, and our gate is its estimator-derived admission rule; gating *vanilla* RLVR updates by saturation signals has independent precedent (5’s Bayesian boundary gating), not applied to relabel destinations. So the precise claims we retain: no prior work (i) derives the sampling utility from the estimator’s own expected signal, (ii) couples it with success-conditioned weights it provably matches, or (iii) *measures the coverage cost of hindsight relabeling in RLVR* and gates relabel admission by destination saturation.

8 Limitations and negative results

Scale. The LLM-scale teacher×estimator interaction is 1-of-2 seeds and tracks steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7). The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. The maze coverage divergence (nine runs, zero sign exceptions, but a heterogeneous pool — §6.3) is the strongest evidence this paper has for the safety claim, and it is exploratory; the balanced maze factorial and a steering-controlled multi-seed LLM cell are the decisive next experiments.

Where the teacher cannot work as specified. Two structural limits at production scale, stated as design constraints rather than future work. *Posterior starvation is the generic regime*: a per-prompt Beta posterior needs multiple visits, but realistic RLVR pools see ~ 1 epoch ($\leq 2\text{--}3$ visits per prompt), so FrontierMax as specified is a small-pool instrument; deploying it on 10^5 -prompt pools requires coarser state — bucket-level posteriors, a learned difficulty predictor, or warm starts — and our external annotations transferred poorly ($\rho = -0.17$), so cold-start from off-model labels is not the answer. *The throughput mechanism does not transfer*: at matched wall-clock the maze teacher’s measured value is extra optimization steps from skipping dead groups’ backward passes, but in generation-dominated LLM pipelines (85% of step time) a dead group has paid its full cost before it is known dead — the relevant LLM-scale comparisons are generation-budget-matched (resample-to-fill, prompt pre-filtering), and the teacher’s value there rests on avoiding the generation itself, which requires the posterior it starves for. These two constraints compound, and we state them together because they bound the same deployment.

Scope of the surrogate. Coefficient mass is not gradient norm, SNR, or expected improvement (Remark 1) — it omits score-vector geometry, sequence length, token masking, optimizer state, and clipping — and §6.5 exhibits the regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment’s relabel map; noisier verifiers will land between our fixed-pool and one-shot endpoints (+0.22 vs. +0.01 AUC).

The coverage main effect is pool-conditional. A pre-registered fifth suite (Jugs water-measuring puzzles; preregistration, per-cell raw endpoints, verdicts, and postmortem vendored under `jugs_llm/results/` with execution-commit provenance) returned all-null: on a pool whose learnable band is a single stratum of 1–2-move template solutions, *plain MaxRL* also collapses — entropy $1.36 \rightarrow 0.07\text{--}0.21$ in 60 steps, frontier-tier pass@16 $0.15 \rightarrow 0$, every arm converging to the same deterministic policy — and recycling accelerates the collapse rather than rescuing it. When N rollouts of a near-trivial plan are near-identical, the i.i.d.-rollout diversity the group contrast requires is absent, and success-conditioned weights liquidate frontier coverage through shared parameters just as GRPO’s do elsewhere. The estimator main effect of §6.3–6.4 is therefore a claim about pools with a graded band at the deployed N , not an unconditional safety guarantee; per-task rollout-set diversity at initialization is the design gate we now check first. The same suite also exposed a granularity bug in the gate (keying destinations by value collides tasks when achieved values are few and shared — ≤ 20 integers in Jugs vs. hundreds in Countdown); the corrected gate keys on the relabeled task.

What did not transfer. A $4\times$ duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ -concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T ; and feeding relabeled successes to the teacher’s posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a “beats the oracle” comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected match-within-noise). Full postmortems live in the repository’s audit trail.

9 Conclusion

For the practical success-conditioned estimator under i.i.d. binary rewards, the expected absolute coefficient mass of a prompt is exactly $2(\text{pass}@N - \text{pass}@1)$, and the drop-all-fail implementation targets truncation order $N-1$. That identity supplies a compute-aware sampling heuristic and an admission threshold for verified relabels; it does not by itself prove a learning band, optimal prompt allocation, or on-policy hindsight, and we have been explicit about where each stronger reading fails. Empirically, estimator choice and relabel admission changed the trade between mean accuracy and $\text{pass}@k$ coverage everywhere we looked: an exploratory but zero-exception coverage divergence between MaxRL- and GRPO-labeled maze runs, a three-seed replication of recycling-induced sharpening on Countdown with one gated operating point that buys the coverage back, and a pre-registered negative (Jugs) marking where the whole family of interventions fails together. The confirmatory experiments these results specify — a balanced estimator-by-sampler factorial at six or more seed blocks, a fixed-code gate-strength sweep, and a steering-controlled multi-seed LLM cell — are pre-registered and queued. The portable practice we advocate is independent of our method: measure coverage next to mean accuracy whenever a curriculum or recycler ships, conditioned on the estimator it ships on.

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A Full details for compressed experiment rungs

Material moved from §6 during the page-budget pass; every number here is artifact-backed in the repository.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr $\times$ 2 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ -matched oracle 0.8885 $\pm$.0014, full stack 0.8895 $\pm$.0023, oracle+recycling 0.8935 $\pm$.0012. The relabel direction term (+.037 over replay) has seed spread $\pm$.0025, 4–6 $\times$ tighter than every placebo; its worst seed beats every placebo’s best. Random-target relabels are harmful relative to dose-matched replay (−.032, 5/5 seeds) but neutral vs. plain lr-doubling: fake labels waste the slot rather than poison training. Utility-form controls: exact u_N 0.728 $\pm$.002 vs. legacy frontier form 0.733 $\pm$.013 (the derived form’s payoff is 7 $\times$ smaller seed variance); the $N=2$ slice $p(1-p)$ is a wash against uniform (−.007, 2/5 wins) where u_N carries +.078 (5/5). AUC spread across arms is 9 $\times$ the final-performance spread.

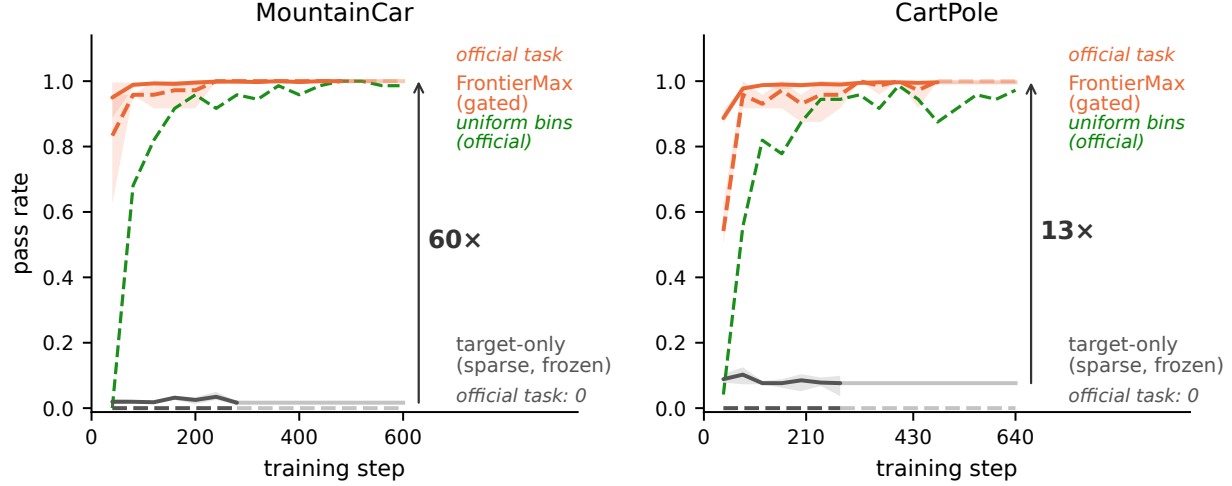


Figure 8: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward; a uniform-over-bins control (green) also cracks the task, 3× slower and off-ceiling; the gated FrontierMax stack converges to 1.000 ± 0.000 .

6.5 IsaacLab protocol. Five arms: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed, pre-registered protocol. The teacher’s only edge over the greedy walker is a within-noise sliver at the hardest evaluated level.

6.6 gym control (full). MountainCar flag: measured random-policy success $< 3 \times 10^{-6}$ over 10^6 episodes; survive-400 $\approx 10^{-13}$ by tail fit. Dense-reward controls (prediction registered before the run): CartPole’s native +1/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.96 ± 0.03 , 3 seeds); MountainCar’s native −1/step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 → +recycling .895 → +teacher .956; CartPole .708 → .792 → .893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect 3.0× to hard-task 0.9 ($2.1 \times$ to 0.99 on CartPole). The per-bin ablation replicates the 10-seed transition-matched study (flag 0.848 ± 0.058) before strengthening it to full convergence.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated; artifact `maze_gpu/efficiency.json`; Fig. 9 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k \approx 4$; 5.12 vs. 4.62 at $k=64$, with GRPO’s level-2/4 curves saturating at 0.88/0.56 where MaxRL’s reach 1.00/0.62.

6.9 secondary reads (full). Training-side damage: ungated recycling ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired +.083/ +.063/ +.034). Entropy at 3 seeds: the gate’s retention replicates (+.072 paired over baseline, all seeds positive, $\sim 1.6 \times$); ungated recycling’s endpoint entropy is seed-unstable (spread ± 0.062 , 4–9× the other arms’), consistent with its noisy frontier-coverage endpoints.

6.7 posterior diagnostics and harness caveats (full). The pass@ k sweep’s absolute level sits $\sim 3 \times$ below the trainer’s val metric (unreconciled harness discrepancy; only the within-sweep contrast is quoted).

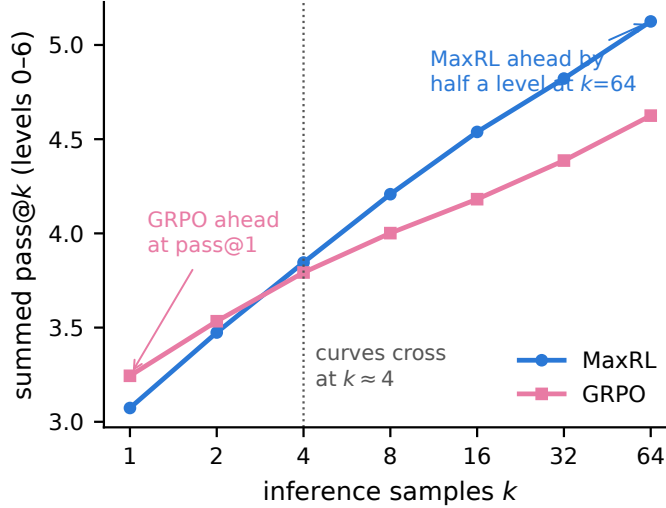


Figure 9: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** GRPO leads at pass@1; the curves cross at $k \approx 4$; MaxRL leads by half a level at $k=64$ — choose the estimator by how many samples you will draw at inference.

Posterior detail: two independent teacher runs’ posteriors agree with each other (cross-run Spearman $\approx .45$ on co-visited prompts, $n \approx 900$; $\approx .67$ self-consistency with a shared latent) vs. $.10-.22$ against external 7B-model annotations, from ~ 1.25 visits per prompt. Across external-difficulty quintiles spanning solve rates $.33 \rightarrow .98$, mean $\hat{p}$ moves only $.076 \rightarrow .103$ — the compressed landscape that starves allocation. Response length separates runs by estimator, perfectly: all three GRPO runs compress to 209.6 ± 1.8 tokens while both MaxRL runs sit at 232.4 ± 5.7 (common ~ 365 -token warmstart; the teacher moves length ≤ 3 tokens) — sharpening visible in a currency where token entropy is ambiguous. Prompt-uniqueness figures for the replacement confound: $\sim 34\%$ unique prompts seen by teacher arms vs. $\sim 43\%$ uniform.

Relabeling in a PPO-family loop: the wiring contract. For adopters (the details a config cannot express). *Placement*: relabeling runs after reward computation and **before** `compute_log_prob` — old log-probs must be computed under the *rewritten* conditioning, making the relabeled group an on-policy ML term with the estimator’s own weights, not a policy-gradient ratio against stale context (our gridworld ablation priced the stale-context mistake at -0.06 AUC, worse than no hindsight). *Rewrite*: the prompt’s goal slot and the response’s goal mentions are both rewritten (token IDs, attention mask, and position IDs rebuilt; 2-D RoPE only — 3-D mrope is refused loudly); the answer artifact is untouched by construction. *Group semantics*: each parseable failure relabels to its own achieved goal and keeps its group id, so one group can mix relabeled tasks; unparseable rollouts stay failures, preserving contrast ($K < N$). The group contrast is then “reached some verified goal” vs. “reached none” — coarser than 1-of- N of a single task, pre-registered as a deviation, with the verl-normalized advantage verified equal to $N \times$ the reference weights in direction. *Hygiene*: the curriculum posterior observes pre-relabel scores only; no KL-to- reference is applied to relabeled groups in our runs (`kl_coef=0`). Reference implementation in the repository (`verl/utils/hindsight.py`, ~ 270 lines, with CPU integration tests).

Jugs suite (pool-conditionality; §8). Tier grid t0–t4 (2–5 jugs, disjoint min-moves bands, 200 unique tasks/tier); feasibility landscape (SmoLM2-360M two-shot, $N=16$): t0 $.058/.475$ (pass@1/pass@16), t1 $.006/.100$, t2–t4 0/0 with relabel-yield-on-failure $.62-.73$ on every tier. All nine pre-registered B-arm cells (no-recycling / ungated / gated $\times 3$ seeds) converged to the same deterministic fixed point: t0 $\text{mean@16} = \text{pass@16} = 0.450$ exactly (the 2-move template stratum), t1 pass@16 $0.15 \rightarrow 0$, actor entropy $1.36 \rightarrow 0.02-0.21$. Ungated recycling ran 169 relabels/step at yield $.88$ and ended with the lowest entropy ($0.02-0.03$); the value-keyed gate rejected 99.8% of relabels (the granularity bug described in §8, since fixed and regression-tested). Postmortem and per-cell artifacts in the repository.

B Hyperparameters and compute

Total compute for every experiment in this paper: ~ 230 A10G-hours on a single GPU plus CPU-days for the exact-gradient suites. Step-time telemetry (artifact `generation_timing.json`, 1,188 mixed GSM8K/Countdown steps): step-weighted mean generation fraction 31% overall; the eight generation-dominated sessions ($>80\%$ session mean, 282 steps) average 83% — a selected-subset description, not an estimate of typical cost. Table 2 gives the per-suite configuration and Table 3 the knob inventory with provenance.

Table 2: Per-suite configuration. Teacher knobs (decay, floor, γ) are shared across suites with the defaults below; what varies per suite is scale, not schedule.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
rollouts N	16	32	16	16
groups/step	1	8	64	64
optimizer, lr	SGD, 0.5	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s clock	50 steps	60 steps
seeds (headline)	5–10	3	1–2/cell	3/arm
compute/run	CPU-min	0.7 A10G-h	~ 20 A10G-h	~ 6 A10G-h

Table 3: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
peak location/ N -scaling	—	derived (Prop. 1)	the theory claim
posterior decay	0.7	tuned (V2b)	closes $\sim 19\%$ of oracle gap
uniform floor	0.1	tuned	0.15 on maze; robust 0.05–0.2
γ (concentration)	1	tuned (V6)	4 on chains; no transfer to maze
<code>gate_max_p</code>	0.5	$p \rightarrow 1$ zero, midpoint tuned	one trade point validated (§6.9)
relabel dose cap	12 groups/step	tuned	sets <i>where</i> coverage is spent (§6.4)

Reproducibility

Every proposition has an enumeration/Monte-Carlo verification script (`test_mass_formulas.py` in `curriculum_maxrl/` checks the MaxRL, RLOO, and both GRPO-SD mass formulas against exact enumeration and the deployed code). Every figure regenerates from committed inputs listed, with checksums and commands, in the frozen manifest `paper/results/manifest.json`: result values live in versioned JSON tables under `paper/figures/data/` (none are hard-coded in figure scripts), and the maze cohort of Fig. 5 is frozen in an explicit run manifest — adding a run file cannot silently change the figure, and missing inputs are a hard error. The Jugs preregistration, raw cells, verdicts, and postmortem are vendored in `jugs_llm/results/` with preregistration- and execution-commit provenance. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team’s summary (raw logs live with that team); per-step timing fractions come from verl’s in-run telemetry (`generation_timing.json`); the fig2/fig3 endpoint tables are transcribed from run analyses rather than derived from a seed-level results table, and building that table is queued. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.